

CONTENTS

CHAPTER 1. DEFINATION AND CONCEPTS OF AGROFORESTRY	3
1.1. Definition and Concepts of Agroforestry	3
1.2. Advantages of Agroforestry	5
CHAPTER 2. CLASSIFICATION OF AGROFORESTRY SYSTEM	6
2.1. Structural Classification of Agroforestry System	7
2.1.1. Classification based on the nature of components	7
2.1.2. Classification based on the arrangement of components	7
2.1.3. Classification based on function of the system	8
2.1.4. Ecological classification	8
2.1.5. Classification based on socioeconomic	8
CHAPTER 3. AGROFORESTRY PRACTICES	9
3.1. Shifting cultivation	10
3.2. Improved tree fallow	12
3.3. Homegardens	13
3.3.1. Function of homegarden	13
3.3.2. Food production from homegarden	13
3.4. Alley cropping	14
3.5.3. Species used in alley cropping	14
3.5.4. Establishment of hedgerows	15
3.6. Trees on Crop Lands	15
CHAPTER 4. AGROFORESTRY SPECIES	16
4.1. Multipurpose Tree Species	16
4.2. The best multipurpose trees for agroforestry	17
4.3. The role of multipurpose trees (MPT) in AF	18
4.4-Role of MPT in different agroforestry practices	19
4.4.1. Role of MPTs in alley cropping	19
4.4.2. Role of MPT in Home gardens and other multistory systems	19
4.4.3. MPTs used as a living fences	20
4.4.4. MPTs used for windbreaks	20
4.4.5. MPTs used in improved fallow systems	21
4.4.6. MPTs in the silvopastoral system	21
CHAPTER 5. COMPONENT INTERACTIONS	23
5.1. Different approaches of understanding of component interaction	23
5.2 Negative and positive component interaction	25
5.2.1. Positive (Production-enhancing) interactions	25

5.2.2. Negative (production-decreasing) interactions	26
5.2. Component management	30
CHAPTER 6. AGROFORESTRY AND SOILS	33
6.1. Beneficial effects of trees on soils	33
CHAPTER 7. AGROFORESTRY POLICES FOR SUSTAINABLE LAND USE	38
7.1. Benefits of Agroforestry that Attracts the Attention of Policy Makers	38
7.1.1. Benefits of agroforestry	38
7.2. Policy Measures to Encourage Appropriate Agroforestry	40
7.2.1. Policy and Implementation Issues	40
7.2.2. Fiscal (Tax and Subsidy) Policy Measures	42
CHAPTER 8. DIAGNOSIS AND DESIGN METHOD	42
8.1. Definition and Concepts	42
8.2. Diagnosis and Design implementation	43
8.2.1. Analysis of Land Use System and Constraints	44
8.2.2. Description of Land Use Systems (LUS)	45
8.3. Basic procedures in conducting D and D	46

CHAPTER 1. DEFINATION AND CONCEPTS OF AGROFORESTRY

1.1. Definition and Concepts of Agroforestry

A strictly scientific definition of agroforestry should stress two characteristics common to all forms of agroforestry and separate them from the other forms of land use, namely:

- The deliberate growing of woody perennials on the same unit of land as agricultural crops and/or animals, either in some form of special mixture or temporal sequence
- There must be significant interaction (positive and negative) between the woody and non woody components of the system, either ecological and economical.

These ideas were refined and the following definition of agroforestry was suggested:

Agro-forestry is a collective name for land use systems in which woody perennials (trees, shrubs, palms, bamboos etc) are deliberately grown on the same piece of land as agricultural crops and/or animals, either in some form of spatial arrangement or temporal sequence. In agro-forestry systems, the woody component interacts ecologically and economically with the crop and/or animal components. Such interactions will take many different forms, both positive and negative, and they need not remain stable over time.

From the definition of Agroforestry we can derive the following points:

- a) AF normally involves two or more species of plants (or plant and animals), at least one of which is woody perennial,
- b) An AF system always has two or more outputs
- c) The cycle of an agroforestry system is always more than one year,
- d) Even the simplest agroforestry system is more complex, ecologically (structurally and functionally) and economically than monocropping system.

Agroforestry is considered to be an interface between agriculture and forestry and encompasses mixed land use practices. AF practices have been developed primarily in response to the special needs and conditions of tropical developing countries that have not been addressed by advances in conventional agriculture or forestry.

Generally, the aim and rationale of most agro-forestry systems are to optimize the positive interactions in order to obtain a higher total, a more diversified and/or a more sustainable production from the available resources than is possible with other forms of land use under prevailing ecological, technological and socioeconomic conditions.

Agroforestry can be either Intentional or Interactive.

Interactive Agroforestry is designed to minimize negative and maximize positive interactions between trees, other crops, livestock and humans. The goal is to enhance the production of more than one component at a time while providing for environmental benefits.

Intentional Agroforestry is designing and managing combinations for a planned result. Intentional Agroforestry can be intensive or integrated. In the practice of Intensive Agroforestry, components are managed to maintain production and environmental benefits. Integrated Agroforestry is a blend of agriculture, forestry and environmental science.

There are three attributes (qualities) of all agroforestry systems

1. Productivity

Most, if not all, agroforestry systems aim to maintain or increase production (of preferred products) as well as productivity (of the land). Agroforestry can improve productivity in many different ways. This include: increases output of tree products, improved yield of associated crops, and reduction of cropping system inputs, and labour efficiency.

2. Sustainability

Sustainability means stability and maintained productivity of a system over a long period of time (indefinitely). Opposite, unsustainable means that the resource base deteriorates and productivity will at a certain time decline, maybe as a result of a short-sighted benefit. Sustainability in small subsistence farming system is crucial because farmers totally rely on limited resources and cannot easily shift to other livelihood. Agroforestry is an integrated system, in which trees and food / cash crops are grown together within the same limited system. Domestic animals are often, but not always, included. Trees, crops and animals have different roles in the system; the balance and interaction between them is the most important aspect of system management, as it both attempt to optimize productivity and assure long term sustainability.

Generally by conserving the production potential of the resource base, mainly through the beneficial effects of woody perennials on soil, agroforestry can achieve and indefinitely maintain conservation and fertility goals. Agroforestry is a sustainable land use system. The essential feature of sustainable land use system is an increase in production with conservation of natural resources on which production depends.

3. Adoptability

The word “adopt” here means “accept,” it may be distinguished from another commonly used word “adapt”, which implies “modify” or “change”. The fact that agroforestry is a relatively new word for an old set of practices means that, in some cases, agroforestry has already been accepted by the farming community. However, the implication here is that improved and new agroforestry technologies that are introduced into new areas should also conform to local farming practices.

1.2. Advantages of Agroforestry

It is obvious that agroforestry is managed for multiple advantages compared to other land use systems. The farmers benefit in many ways with a well-designed, well planned and well executed agroforestry system. The presence of trees and shrubs aid the crops in making better utilization of the soil nutrients and light, hence resulting in better production as compared to monocrop method. Enhancement in the nutrient cycling capabilities of the soil reduces the cost for weed and pest control. Agroforestry systems provide an option of earning extra income to the farmers. Diversification between forest and agricultural crops reduces labor and resource costs along with the risks undertaken by farmers. Mixing long term forest crops with yearly agricultural income makes the overall system more profitable. Lastly forest provides food and shelter for the livestock which is beneficial in many ways.

Let us summarize the advantages and disadvantages of agroforestry:

Merits	Limitations
✓ Reduction in fallow period and high cropping intensity over longer time period	✓ High labor input
✓ Erosion control and runoff management	✓ Highly skilled management
✓ Strengthening of nutrient cycling mechanisms leading to savings in fertilizer use	✓ Low yields due to allelopathic effect and competition among trees and food crops for light, water, and nutrients

- ✓ Alternate products (e.g., fodder, fuel, staking water mulch and food crops)/Provision of diverse products and poverty reduction
- ✓ Saving, land and decrease in need for clearing new land
- ✓ Improved traditional system, therefore, ecologically compatible
- ✓ Soil fertility and closed nutrient cycles
- ✓ Soil salinity control
- ✓ Stabilization of soils and microclimate
- ✓ Low input of agrochemicals
- ✓ Improvement of wildlife and pollinator habitat
- ✓ Remediation of polluted soils
- ✓ Prevention of damage to forests
- ✓ Climate change mitigation
- ✓ Potentially high risks of pests and diseases
- ✓ Difficulties in adoption due to traditional land tenure system

CHAPTER 2. CLASSIFICATION OF AGROFORESTRY SYSTEM

The main purpose of classification of agroforestry system is to provide a practical framework for the synthesis and analysis of information about existing systems and the development of new and promising one. Therefore any classification scheme should:

- Include a logical way of grouping the major factors on which production of the system will depend
- Indicate how the system is managed- pointing out possibilities for management interventions to improve the system's efficiency
- Offer flexibility in regrouping the information; and
- Be easily understood and readily handled - practical

The most obvious easy to use criterion for classifying agroforestry systems are the spatial and temporal arrangement of components, the importance and role of components, the production aims or out puts from the system, and the social and economic features. They correspond to the systems' structure, function (output), socioeconomic nature, or ecological (environmental spread). These characteristics also represent the main purpose of a classification scheme.

2.1. Structural Classification of Agroforestry System

The structure of the system can be defined in terms of its components and the expected role or function of each, manifested by its outputs. However, it is important to consider the arrangement of components in addition to their type.

2.1.1. Classification based on the nature of components

In agroforestry system there are three basic sets of elements or components that are managed by the land user, namely, the **tree or woody perennial, agricultural crops and the animals**. In most agroforestry systems, the herbaceous species also involved, the notable exception being apiculture and aquaculture with trees, and plantation - crop mixture of two woody perennials such as coffee and rubber tree and tree under shade trees. Animals are only present in some agroforestry systems.

Based on the type of components existing, agroforestry can be grouped in to

1. **Agrisilviculture:** where the two components are trees and agricultural crops. It is a common practice in the lowland humid tropics at a subsistence level of production.
2. **Silvopastoral:** where the two components are trees and animal/pasture. This system is a common practice in pastoral and some agropastoral areas.
3. **Agrosilvoculture:** where the components are trees, agricultural crops and animal/pasture. This system is a common practice for food production and soil conservation in highland humid tropics.

2.1.2. Classification based on the arrangement of components

The arrangement of components refers to the plant components of the system (especially if the system involves plants and animals). Such plant arrangements in multispecies combinations can involve the dimensions of space and time. **Spatial arrangements** of plants in agroforestry mixtures vary from dense mixed stands to sparsely mixed stands. Moreover, the species can be in zones or strips of varying widths. There can be several scales of such zones varying from micro zonal arrangement to macro zonal once. A commonly known example of the zonal arrangement is hedgerow intercropping – alley cropping. An extreme form of zonal planting is the boundary planting of trees on edge of plots and fields for a variety of purposes and outputs (fruit, fodder, fuelwood, fencing, soil conservation...). It is also important to note that extreme forms of macrozonal arrangement can also be construed as sole cropping system, the

interactive association of different components, however, can be used as the criterion to decide the limits between macrozonal agroforestry and sole crop system.

Temporal arrangement of plants in agroforestry can also take various forms. An extreme case is the conventional shifting cultivation, when selected woody species or mixture of species is planted or is allowed to regenerate naturally. Temporal arrangement of components in agroforestry have been described by terms such as coincident, concomitant, overlapping, separate, interpolated, and so on.

2.1.3. Classification based on function of the system

Production and protection (which is the cornerstone of sustainability) are, theoretically, two fundamental attributes of all agroforestry systems. Agroforestry has both productive and service functions. The distinctive contribution to production is to obtain tree products from the farm; these include the fuel wood, fodder and fruit, together with construction wood and a host of other items, such as gums, resins and medicinal products. The relative importance of these products varies between systems, according to environment and socioeconomic circumstances. The range of products in agroforestry serves to diversify the output from farms, giving a broader economic base and greater food security (FAO, 1989). The service function also varied, including shade (for human and livestock), reduction in wind speed, control of weed and fencing. There is no doubt, however, that the major service function of agroforestry is its role in soil management, including control of erosion and maintenance and improvement of soil fertility.

2.1.4. Ecological classification

Most agroforestry categories can be found in all agroecological zones; therefore, agroecological zonation alone cannot be taken as a satisfactory basis for classification of agroforestry systems. However, agroecological characteristics can be used as a basis for designing agroforestry systems, because, similar agroecological regions can be found in different geographical regions, and the agroforestry system in similar ecological zones in different geographical regions are structurally (in terms of nature of species composition) similar. The main point is that several types of agroforestry systems and practices are relevant to any major agroecological zone; depending on the special conditions of the zone, the emphasis of the system or practice will also vary.

2.1.5. Classification based on socioeconomic

Socioeconomic criteria such as scale of production and level of technology input and management have also been used as a basis for classifying agroforestry systems. Lundergren (1982), for

example grouped systems into **commercial, intermediate and subsistence** systems. As to Lundergren, the **commercial type of agroforestry system** is focused on production of out puts (usually a single commodity) for sale. In this system, the scale of operation is often medium to large and land ownership may be government, companies or private; labour is normally paid or otherwise contracted. Examples of such kind of agroforestry system are production of plantation crops (Coconut, Oil palm, rubber...) with permanent understories of food crops.

The **intermediate type of agroforestry systems**, on the other hand, is those that are intermediate between commercial and subsistence scale of production and management, i.e. production of perennial cash crops and subsistence crops undertaken on medium to small sized farms where the cash crops cash needs, and the food crops meets the family's food needs. Usually farmers who either own the land, or have long term tenancy rights to land, reside and work on the land distinguishing the intermediate system from the commercial system at one end and from subsistence system on the other, are holding size, and level of economic prosperity.

The **subsistence agroforestry systems** are those where the use of land is directed towards satisfying basic needs and is managed by the owner or occupant and his/her family. Anthropologists define subsistent farmers as those who produce most of what they consume, or consume most of what they produce. Farmers who do not, or can not, produce enough for the needs of their families are also usually considered under this category.

Therefore, socioeconomic factors that are likely to change with time and management conditions cannot be rigidly adopted as a satisfactory basis for an objective classification scheme, but they can be employed as a basis for a defined objective or action plan.

CHAPTER 3. AGROFORESTRY PRACTICES

The type of agroforestry techniques that are practiced depends on factors such as the region, type of crops, soil, climatic conditions etc. A wide selection of tree species and woody shrubs can be used for agroforestry systems. Some of these trees are suited for acid soil conditions and others for erosion control and some are more appropriate as forage trees. The choice of appropriate species is critical to the success of agroforestry system. In addition to the intended use, the choice of tree and associated crop species also depends on cultural and ethnic factors, which is of social importance. Some of the most popular agroforestry practices adopted worldwide are outlined below.

3.1. Shifting cultivation

Shifting cultivation or slash and burn agriculture is one of the most ancient agricultural systems, still practiced among millions of farmers in the tropics. Shifting cultivation is a nomadic or semi-nomadic cultivation method and depends as such on available ‘free’ forest resources or implicit: it exists only where such sources are available. Shifting cultivation is still the mainstay of traditional farming systems over vast areas of the tropics and subtropics. Although shifting cultivation is dominant mainly in sparsely populated and lesser developed areas, where technological inputs for advanced agriculture such as fertilizers and farm machinery are not available, it is found in most of the tropics, especially in the humid and sub humid tropics of Africa and Latin America.

Generally, the rationale of shifting cultivation is to utilize accumulated nutrients in the forest biomass for cultivation of agricultural crops. The nutrients are released by burning, which also creates glades with open space and light. The nature of the forest before shifting cultivation is practiced depends on culture, environment, technical development and land pressure. For example, clearing of large trees is a technical problem with less advanced tools; land pressure tends to push shifting cultivation practice from primary to secondary forests.

Shifting cultivation is farming or agricultural system in which land under natural vegetation is cleared, cropped with agricultural crops for a few years, and then left untended while the natural vegetation regenerates.

In shifting cultivation the cultivation phase is usually short (2-3 years), but the regeneration phase known as the fallow or bush fallow phase, is much longer (traditionally 10-20 years). Shifting cultivation always depends on the use of fire for clearing. Burning is done during the dry season. Nowadays these fires often get out of control and cause vast forest fires. Shifting cultivation is practiced on a site as long as there are nutrient enough to support a crop. Fertility is highest right after burning and wears off after a shorter or longer period of time of fire. Cropping period is not sharp. High demanding crops like yams, maize and dry rice are the first crops followed by less demanding crops like cassava and bananas. Plots often continue be used extensively for collection several years after cropping has been abandoned.

Shifting cultivation is typically a subsistence farming system. It is practiced by both native tribes and immigrants, the latter often landless poor from marginal areas. Shifting cultivation is space demanding as the method depends on an interim period of natural regeneration of the forest. It occurs thus mainly where population pressure is low, which implicitly also are where soil fertility is too low to support a permanent sedentary agriculture culture. The practice has turned widespread after commercial logging of natural forest. Most Shifting cultivation practices occur in the humid or semi-humid tropics and tropical highland, where above conditions prevails and where regeneration is reasonably fast as compared to dryer tropics.

In almost all classification schemes, the various categories designate different degrees of intensification of cultivation which can best be evaluated on the basis of the land use factor (L):

$$L = \frac{C + F}{C}$$

C = length of the cropping phase (Years)
F = length of the fallow phase (years)

During the early stages of shifting cultivation, when fallow periods are long,

$L > 10$. However, when a sedentary and permanent cultivation stage is reached, as on the compound farm, $L = 1$. Moreover, the various systems of shifting cultivation are interwoven in the agricultural landscape. This is particularly so in Africa where one can find traditional shifting cultivation and permanent production systems existing together in the same locality. Thus, within the general pattern of alternating fallow and cropping cycle, the nature of shifting cultivation varies from place to place.

Large parts of humid and sub humid tropics currently under shifting cultivation and related traditional farming system are covered by the so called fragile upland soils. It is generally accepted that traditional shifting cultivation with adequately long fallow periods is a sound method of soil management, well adapted to the local ecology and social environment. Before the soil is cleared, a closed nutrient cycling exists in the soil–forest ecosystem. Within this system, most nutrients are stored in the biomass and topsoil, and a constant cycle of nutrients transfer from one compartment of the system to another operates through the physical and biological processes of rain-wash, litter fall, root decomposition, and plant uptake.

Clearing and burning the vegetation leads to a disruption of this closed nutrient cycle. During the burning operation the soil temperature increases, and afterwards, more solar radiation falling on the bare soil surface results in higher soil and air temperature. This change in the temperature

regime causes change in the biological activity in the soil. The addition of ash to the soil through burning causes important changes in the soil chemical properties and organic matter content. In general exchangeable bases and available phosphorus increase slightly after burning; pH values also increase, but usually temporarily. These changes in the soil after clearing and burning result in an increase of available nutrients, so that the first crop that is planted benefits considerably. Afterwards, the soil becomes less and less productive and crop yield decline.

3.2. Improved tree fallow

The practice of leaving cropped fields to lie fallow in order to allow the soil to rest and recover some of its fertility is well known and widely used throughout the world. When agroforestry is added, woody species replace crops on the fallow fields in sequence over time. The main feature, which distinguish fallow based agroforestry systems from other agroforestry practices is that trees and shrubs are not grown with crops with the same plot at the same time. In many parts of Africa, fallow periods are becoming shorter, due to an increasingly acute land shortage. The shorter rotations may fail to restore the soil sufficiently to sustain later crop production. As the fallow period is reduced, it became more important to introduce trees and shrubs that could help speed up soil recovery.

An improved tree fallow is a rotational system that uses preferred tree species as the fallow species (as opposed to colonization by natural vegetation), in rotation with cultivated crops as in traditional shifting cultivation.

Improved fallows can be established in a variety of ways, and at various stages of the fallow. Method might include:

- Direct seeding of clean-tilled, harvested plots
- Selective cutting of bush, followed by enrichment planting with tall seedlings
- Introducing tall seedlings and cutting into poor quality fallows on degraded land
- Planting tree seedlings into closely spaced, deep planting holes or furrows within blocks of cleared cropland.

The exact techniques vary with previous land use, value of the fallow vegetation, condition of the land and expected duration of the fallow.

3.3. Homegardens

Homegardens refers to intimate association of multipurpose trees and shrubs with annual and perennial crops and, invariably livestock with in the compounds of individual houses, with the whole crop-tree-animal unit being managed by family labour (Fernandes and Naire, 1986).

3.3.1. Function of homegarden

The various food crops from the homegardens may provide a substantial proportion of the calorific and nutritive requirement of the local diet. Food production is, thus, the primary function and the role of most, if not all, of the homegardens and a large proportion of the production from the homegarden is consumed by the gardeners themselves. Another aspect of the important role of food production in homegardens is the maintenance of almost continues production through out the year from the unit. The combination of crop with different production cycles and rhythms is such that an uninterrupted supply of food products is maintained. Depending upon the climate and other environmental parameters, there may be some peak and slack seasons for harvesting the various products, but in a general sense, there is something to harvest daily from most homgardens. Most of this production is for home consumption, but any marketable surplus can provide a safeguard against failure of, as well as a security for the interval between the harvests of other agricultural crops of the home garden. The management required for homegarden is relatively small amount of working time of the members of the family.

3.3.2. Food production from homegarden

The magnitude and rate of production, as well as the ease and rhythm of maintenance, of the homgarden system depend on its species composition. Although the choice of species is determined to a large extent by environmental and socioeconomic factors, as well as the dietary habit and market demands of the locality, there is a remarkable similarity with respect to species composition among different homegardens in various places, especially with respect to the herbaceous components. This is so because food production is the predominant role of most herbaceous species, and the presence an over story requires that the species are shade tolerant. Thus, tuber crops such as cassava, sweet potato and yam dominate because they can be grown with relatively little care as understory species in partial shade and yet be expected to yield reasonable level of carbohydrate-rich produce. Harvesting can be staggered over several weeks depending on household needs.

A conspicuous trait of the tree crop component in homegardens is the predominance of fruit trees, and other food producing trees. Apart from providing a steady supply of various types of edible products, these fruit and food trees are also compatible- both biologically and environmentally- with other components of the system. Produce from the trees often provides a substantial proportion of the energy and nutrient requirement of the local diet.

3.4. Alley cropping

Alley cropping, also known as hedgerow intercropping, involves managing rows of woody plants with annual crops planted in alley in between. Or it is growing of food crops between hedgerows of planted shrubs and trees, preferably leguminous species. ***Alley cropping or hedgerow intercropping is growing of crops in-between trees planted in rows.*** The hedgerows are pruned regularly during the crop's growth to provide biomass and to prevent shading of the growing crop.

3.5.3. Species used in alley cropping

Alley cropping usually includes **leguminous trees to improve soil fertility through nitrogen fixation**. Apart from this important trait, some woody species are obviously less compatible with crops than others. Large dense crowns and surface roots extending far beyond the crown line interface with most crops. Trees that use a lot of water may also reduce the water available for the crops. Some trees are poor candidates for any agroforestry practices in crop land because they give off toxic chemicals into the surrounding soil, a defense against weeds and other competitors in the forest environment. Even woody species that are otherwise compatible with crops are often not suited to alley cropping because of their size or poor response to frequent pruning.

An ideal alley cropping tree or shrub should have several characteristics. These are

- It should have a sparse, small crown to permit sunlight penetration or should resprout rapidly after pruning (management practice that removes the lower branches from a tree), coppicing (cutting trees near the ground), pollarding (cutting all branches including the upper part of the stem) or lopping (removal of small branches and the leaves).
- It should form a deep tap root system with few lateral branches near the surface, so as not to compete with crop roots.
- Trees or shrubs may be used with shallow lateral roots that are easily pruned by ploughing along the hedgerow, without serious damage to the plants.
- The leaf litter, or some portion of it, should decompose at a rate that makes nutrients available when they are needed in the cropping cycle.

- Ideally trees or shrubs used for alley cropping should fix nitrogen and should also produce wood, food, fodder, medicine or other products used by farmers or other members of the local community.
- Finally, the species selected should grow well under the specific limitations of the site, such as saline or acid soils, drought, flooding, heavy winds, insect pests or other hazards.

The most commonly introduced species in alley cropping include: *Leucaena Leucocephala*, *Cassia Siamea*, *Gliricidia Sepium*, *Callindra Calothyrsus*, *Sesbania Sesban*, etc. It should be noted, however, that *Leucaena Leucephala* performance varied according to climate and soil type. It has often failed, and even hampered neighboring crops, because it attracts animal and insect pests, including domestic stock, small antelopes and termites. The criteria listed above can only serve as a guide for species selection at particular sites. In some cases, it may be necessary to combine two or more species in the hedgerow to provide the products and benefits desired.

3.5.4. Establishment of hedgerows

Hedgerows may be established by direct seeding or by planting seedlings or cuttings. Seedlings are placed in a deep ploughed furrow or in deep individual planting holes. During establishment, the hedgerow may require protection from browsing animals, trampling or pests. The farm trees will benefit from weeding, fertilizing and whatever management the farmer follows for the crops. These activities will encourage growth of the hedgerows so that later they require little additional attention. Once established, trees and shrubs used in alley cropping are usually left to grow for 6-18 months before the first cutting.

3.6. Trees on Crop Lands

An agroforestry practice that involves trees on croplands consists of trees grown on cropped lands, in an open, mixed spatial system. The trees may be grown on farmer's fields while crops are grown in the under storey. The trees may be dispersed widely, either systematically in a grid or scattered at random. Generally the trees and shrubs may be natural, left when the land was cleared, or artificially planted. They are usually more or less randomly distributed, although a grid (systematic) pattern is possible when planted.

The practice of rising trees dispersed on crop land may be based on

- Protection and management of selected mature trees already on the site,
- It may involve planting new trees or

- It may depend upon careful management of selected seedlings established on site through natural regeneration.

Farmers may have several reasons for growing trees dispersed on cropland. Some of the most common are:

- ✓ To increase crop production
- ✓ To maintain land productivity for longer time
- ✓ To increase the total yield of mixed products from the crop land
- ✓ To diversify the range of products

Trees growing on cropland should compete only minimally or not at all with the crop and if possible, should contribute to the crop growth. The introduced species should have desirable characteristics such as:

- ✓ a deep rooting system,
- ✓ a form that produces only light shade,
- ✓ a capacity to improve the soil through N-fixation and/or leaf litter and
- ✓ No tendency to harbour crop pests.

Besides, farmers also keep trees on cropland to obtain tree products, such as fruit or fuelwood. However, trees may harbour birds or other pests, which can damage field crops. Several species have been suggested for planting on cropland: the most well known is *Acacia albida* (*Faidherbia albida*). There are many other species (both exotic and indigenous): *Terminalia* spp, *Cordia abyssinica*, *Albizia schimperii*, *Gravillea robusta*, *Acacia tortilis*, *A. Senegal*, *Sesbania sesban*, *Croton macrostachyus* etc. The function of these trees/shrubs on croplands is to improve soil productivity and to provide fuelwood and fodder. The trees/shrubs build up the fertility of the soil and contribute to a more stable environment. Soil erosion is reduced. Nitrogen is fixed and is made available to the crops. Many indigenous tree species in Ethiopia mix well with farming activities, particularly with the raising of staple crops. The trees also provide pods and leaves for supplementary livestock feed and a limited amount of branches for thorn fences.

CHAPTER 4. AGROFORESTRY SPECIES

4.1. Multipurpose Tree Species

The multipurpose tree species (MPTs) is a plant species that is purposefully grown so as to provide two or more than two products and also service function like shelter, shade, and land sustainability

of the land-use system. It is well known that all trees are said to be multipurpose (MT.); but some are more multipurpose than others. Farmers have been growing trees for different purposes for thousands of years. Certainly, all trees have at least more than two service and/or productive roles. In AF context, multipurpose trees are defined as: ***"All woody perennials that are purposefully grown to provide more than one significant contribution to the production and/or service functions of a land use system"***.

4.2. The best multipurpose trees for agroforestry

It is important to select the most suitable trees since it is not easy to replace them once they have been planted. The following factors should be kept in mind when selecting tree species.

A-Environmental adaptation

A multipurpose tree must be able to adapt to the area's climate, soil, topography, and plant and animal life. This is especially important for exotic species, that is, species introduced from outside the project area or from outside the country.

B-Needs of farmers

The species should meet the needs of farm families. For this reason, it is important to involve farmers, both men and women, in selecting species. They should identify desired tree products and planting locations based on local markets as well as conservation needs. The cost of acquiring seedlings or cuttings should also be kept in mind. Planting stock of most fruit trees is expensive.

C-Ease of maintenance

Some species are more difficult to look after than others. Farmers should consider beforehand how much time they have to care for the trees. If they require additional skills and knowledge to grow a particular species, training or demonstration programs should be organized.

D-Availability of genetic materials

Seeds or seedlings of the species being considered must be easy to obtain. If vegetative propagation is required, farmers should receive training in how to do this. Seeds of *Gliricidia sepium* and *Sesbania* spp., for example, might not be readily available. Farmers may need help in how to propagate the trees using root cuttings. They may not even have enough time to harvest the cuttings.

Characteristics of MPTs suitable for Agroforestry

- ✓ Adaptability to local climatic conditions.
- ✓ Ease of establishment from seeds and seedlings.
- ✓ They should have a low crown diameter to bole diameter ratio.
- ✓ They should be light branching in their habit.
- ✓ They should tolerate relatively high incidence of pruning.
- ✓ Deep thrusting tap root system and few and shallow lateral roots (or prunable).
- ✓ Their phyllotaxies should permit penetration of the light of the ground.
- ✓ Good Coppicing ability.
- ✓ Effective nutrient recycling.
- ✓ Multiple uses: food, feed, firewood, construction materials and other products and service (shade, shelter etc.)
- ✓ Minimal competition with shallowly rooted annual crops.
- ✓ Small leaflets readily detached when dried and quickly decomposed when used as fertilizer.
- ✓ A high proportion of leaves to secondary branches.
- ✓ Good tolerance for drought, low fertility and others.
- ✓ Freedom from pests and diseases.
- ✓ Not compete for moisture, space and air.
- ✓ Ability to fix atmospheric nitrogen.
- ✓ Supply nitrogen in the soil.
- ✓ Have thin and erect leaves.
- ✓ Maintain proper ecosystem.
- ✓ Non-allelopathic effects on arable crops.
- ✓ Easy to propagate and prolific seed producer.

4.3. The role of multipurpose trees (MPT) in AF

Now a day, there is quite a lot of information available about MPTs that are commonly used in AF. Generally, fodder trees, and fuelwood species, and sometimes fruit trees, are terms that are widely used in AF literature. They represent important group of MPTS.

The following could be a summary of the role of MPTs in agroforestry

- ❖ Trees may provide food, shelter, energy, medicine, and cash income
- ❖ Raw materials for crafts fodder and forage and
- ❖ Resources to meet social obligations.

Trees used in AF system can also provide a variety of services such as being a form of saving and investment, contributing to the improvement of soil fertility for crop production, protecting the soil from erosion and modifying the microclimate.

4.4-Role of MPT in different agroforestry practices

4.4.1. Role of MPTs in alley cropping

Some tree species reduce soil erosion when planted as hedgerows. When planted on slopes, alley crops slow down runoff and trap sediment, which can form natural terraces after several years. If the land slopes, farmers should plant the hedgerows along the contour, that is, the trees in each hedgerow should be planted at the same level of slope. If planted in double lines rather than single lines, hedgerows can produce almost twice as much foliage and wood, without greatly increasing the competition with nearby crops for water, nutrients, and sunlight. Where farmers can plant in a north-south rather than east-west direction, this will reduce competition for sunlight. Whether farmers plant single or double rows depends on how much land is available and the slope.

Desirable characteristics of MPT species used in alley cropping

- ❖ **Easily established:** require minimum labour for planting and maintenance.
- ❖ **Fast growing:** benefits become available to the farm family as soon as possible.
- ❖ **Good sprouting:** hedgerows continue to grow regularly after pruning.
- ❖ **Nitrogen fixing:** leguminous (nitrogen-fixing) species can contribute to crop nutrition.
- ❖ **Heavy and palatable foliage:** provide more green manure and acceptable fodder.
- ❖ **Deep root system:** nutrients and water are drawn from lower soil layers and reduce competition with associated crops.
- ❖ **Easy to propagate:** generally, growing hedgerows from seed requires less labour than vegetative propagation.
- ❖ **Adaptable to close spacing:** hedgerows require dense planting.

4.4.2. Role of MPT in Home gardens and other multistory systems

Many farmers already grow multipurpose trees in their homegardens for flowers, fruits, and seeds. If farmers want to grow a tree for its food rather than its wood or leaves, they should plant seedlings at least 5m from each other. Also, MPTs grown mainly for food should not be pruned regularly for fodder or fuel. Pruning can interfere with flowering and fruiting. Instead, farm families can collect fallen branches for fuelwood.

Desirable characteristics of MPT species used in homegarden

- **Cast the desired amount of shade:** understory crops need the right amount of light for optimum growth. Typically, the crown should be high, small, open, and foliage sparse.
- **Deep rooted:** they can draw nutrients and water from deeper soil layers and will not compete with shallow-rooted crops.
- **Roots should not spread laterally too far from the trunk:** to minimize competition with nearby crops.
- **Nitrogen fixing:** to grow well under adverse conditions and help improve soil fertility.

4.4.3. MPTs used as a living fences

In many places, farmers plant multipurpose trees in rows along farm boundaries as "living fences". In addition to providing fodder and fuelwood, living fences provide privacy and protection from browsing animals.

Desirable characteristics of MPT species used as live fence

- ❖ **Tolerate minor "injuries":** living fences are susceptible to frequent injuries from pruning or animals and should tolerate them well.
- ❖ **Fast growing:** provide benefits to farm families as soon as possible.
- ❖ **Compatible with annual and perennial crops:** should not have adverse effects on other tree species or crops they are associated with.
- ❖ **Provide fodder:** serve as a source of animal feed.
- ❖ **Fire resistant:** act as a fire break.
- ❖ **Thorns:** stiff branches, thorns, spines, nettles, or irritating latex help keep animals out if that is a problem.
- ❖ **Vegetative propagation:** ensures fast establishment while reducing the chance of spreading to pasture and cultivated areas.

4.4.4. MPTs used for windbreaks

Windbreaks are strips of trees, shrubs, and vines planted closely together along the edges of croplands perpendicular to winds. Especially in dry areas, windbreaks can provide protection to crops and soils from the detrimental effects of wind.

Table 4.1. Some examples of MPT commonly used as windbreaks and living fences

Species	Agroecology	Uses other than windbreak
<i>Azadirachta indica</i>	semiarid tropics	timber, lumber, essential oils, fuelwood
<i>Casuarina equisetifolia</i>	humid tropics	fuelwood, timber
<i>Eucalyptus camaldulensis</i>	humid tropics	fuelwood, timber
<i>Leucaena leucocephala</i>	humid subtropics, humid tropics	fuelwood, poles timber fodder
<i>Grevillea robusta</i>	subhumid tropics, humid tropics	timber, fuelwood, building materials

Desirable characteristics of MPTs used for windbreak.

- ✓ **Wind resistant:** withstand strong winds.
- ✓ **Deep spreading root system:** adds stability to the windbreak by making the trees less susceptible to wind damage.
- ✓ **Small open crown:** reduces the risk of wind damage.
- ✓ **Easy to propagate:** minimizes labour inputs.

4.4.5. MPTs used in improved fallow systems

In many regions, population increase has resulted in faster rotation cycles of crop cultivation and shorter fallow periods. In some instances, fallow time is simply too short to allow the soil to recover. Blocks of fast-growing trees, particularly species that fix nitrogen in the soil, can help the soil recover as well as provide fuel, poles and fodder.

Desirable characteristics of MPT in improved fallow system

- **High nitrogen content in tissue:** hastens soils rejuvenation, since nitrogen is the most important deficiency of tropical soils.
- **Fast biomass production:** litter from the tree, especially foliage, can add important nutrients to the soil.

4.4.6. MPTs in the silvopastoral system

Farm systems that combine tree plantations with livestock that graze beneath them are found generally in drier areas, where natural grasslands and farm sizes are larger. In such systems, farmers graze sheep or cattle on forage grasses or in stands of trees. Advantages to integrating tree growing with livestock production include:

- ❖ Increasing the production of meat protein without sacrificing large tracts of agricultural land.
- ❖ Reducing surface soil erosion by preventing open grazing and reducing dependence on grasses.
- ❖ Using animal manure to fertilize the soils and intercrops, reducing reliance on inorganic fertilizer.
- ❖ Providing additional income through sale of livestock.

Desirable characteristics MPTs used in silvopastoral system

- **High protein and nutrient content:** improve livestock nutrition.
- **Palatable:** livestock are selective about what fodder they will eat.
- **Free of toxic substances:** some fodder species contain toxic substances that can be consumed only in small amounts. *Leucaena* spp, for example, contain mimosine which, in large doses, is harmful to animals. It must, therefore, be used in combination with other fodder sources.

Productive and service roles of MPT Species

The productive and service roles of MPT species are summarized in to nine main categories.

Food and nutrition -Increased amounts of food -A year round supply of food -Better - quality food	Energy -Increased fuelwood supply -Better-quality fuelwood -Cheaper and more convenient fuelwood source
Shelter, structure -Building materials - Shade - Protection from wind and Animals -Marking of Boundaries	Medicine -Preventive (to maintain health) -Curative (to treat disease or injuries) -Veterinary medicine
Raw materials for crafts and cottage industry -An increased supply of materials -New materials	Cash income, savings and investment -Employment (cash earnings) -Sale of products (cash earnings) -Substitution of own products for purchased items (less cash spent) -Exchange of products for other goods (less cash spent) -New form of saving & investment -Greater profitability or security of existing saving and investment

Fodder and forage -Primary feed -Supplementary feed	Resources to melt social obligations -New or improved source of support for social obligations
Conservation of soil, water and plant resources -Increased amounts of water for plant growth, domestic use and livestock -Improved seasonal availability of water -Improved amount, quality or timing of water delivery to dams and large scale water works. -Protection of the soil from erosion and loss of nutrients -Restoration of degraded soils -Improvement of soil moisture and fertility -Maintenance or increase in species and habitat diversity -Substitution of farm tree products for over use of woodlands for fuel and other items -Improved conditions for natural regeneration of most desirable species	

CHAPTER 5. COMPONENT INTERACTIONS

5.1. Different approaches of understanding of component interaction

Component interaction is always included in the definition of agroforestry. However, scientists, in different fields of study have used different approaches to express this interaction. Historically, different groups of scientist have described these interactions differently:

5.1.1 Ecologists classification

In the ecological literature, the types of interactions in two-species populations have often been described (Hart, 1974; Trenbath, 1976; Pianka, 1988) on the basis of **net effect** of interactions, by such terms as:

Commensalistic: (positive, “+”, effect on species one and no observable effect, “0”, on species two), or one species benefited without negatively affecting the other species,

Amensalistic: (-, 0), Negative effect on species one and no observable effect on species two, or one species negatively affected but the other species remain unaffected.

Monopolistic, predatory or parasitic: (+, -), One component benefit at the expense of the other,

Inhibitory: (-, -) The two components are negatively interacting or the two components cannot live together, and

Synergistic: (+, +) could be added as an interaction where the net effects are positive for both species.

These concepts of observable net effects can also be expressed by terms such as complementary, supplementary, and competitive, as depicted in next figure; they are used to describe economic interactions as well. **Component interactions refer to the influence of one component of a system on the performance of the other components as well as the system as a whole.**

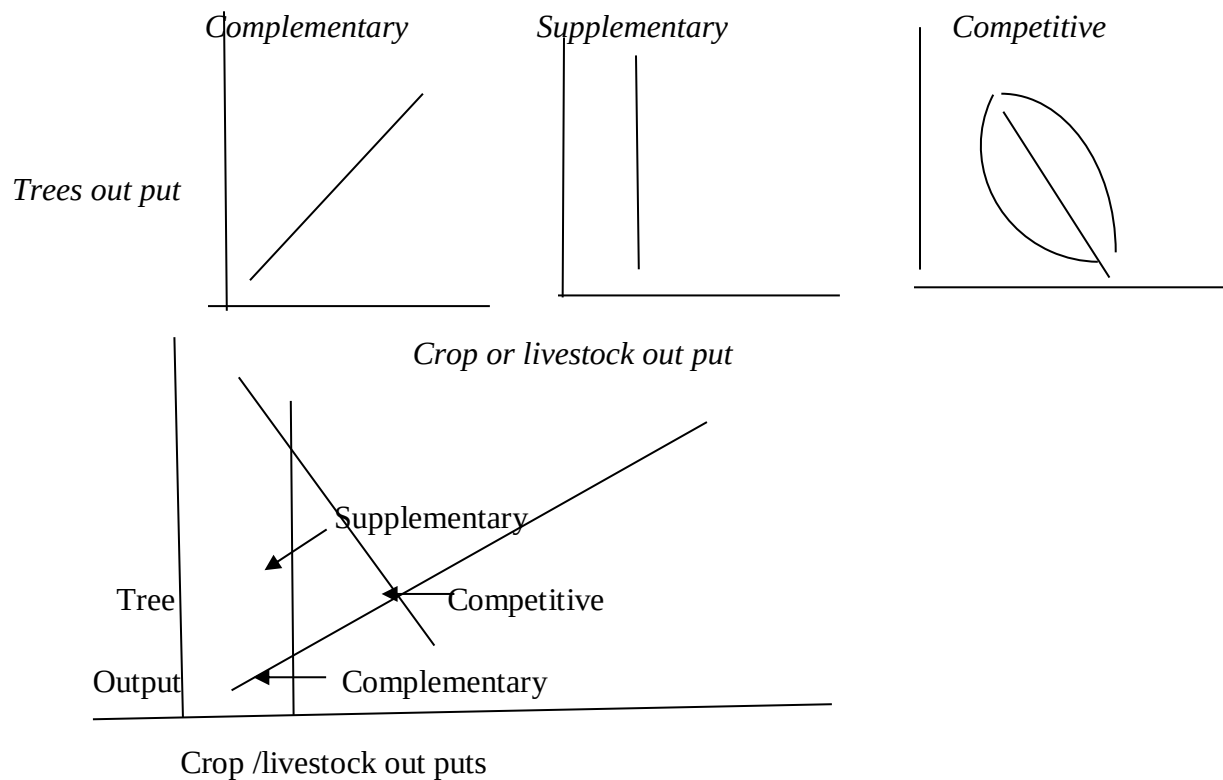


Table 5.1 The major positive and negative effects at the tree-crop interface (TCI) and the tree-animal interface (TAI).

Type of Interaction	At the TCI	At the TAI
Positive	-Shading trees (stress reduction) -Biomass contributions -Water conservation -Soil conservation	-Shading -Manure deposition
Negative	-Light competition -Nutrient competition -Water competition -Allelopathy	-Phytotoxicity -Browsing damage -Trampling -Disease/pest host

5.2 Negative and positive component interaction

5.2.1. Positive (Production-enhancing) interactions

This portion deals not only with the beneficial effects of one component on another, but also on the manipulation of negative effects to minimize their influence on the productivity of the overall system. Let us now discuss the positive interactions at tree- crop interface and tree animal interface separately.

5.2.1.1 At the tree-crop interface

The major types of positive or complementary interactions at the tree-crop interface (TCI) are those relating to microclimate amelioration and nutrient balance. Interactions involving nutrient relations in agroforestry systems are discussed elsewhere in this module (section 5.1); therefore, discussions here will be limited to the other major factor, microclimate amelioration.

In agroforestry systems, microclimate amelioration involving soil moisture and soil temperature relations results primarily from the use of trees for shade, or as live supports, live fences, or windbreaks and shelterbelts. The provision of shade causes a net effect of complex interactions, which extend far beyond the mere reduction of heat and light (Willy, 1975). Temperature, humidity, and movement of air, as well as temperature and moisture of the soil, directly affect photosynthesis, transpiration, and the energy balance of associated crops (Rosenberg *et al.*, 1983), the net effect of which may translate into increased yield. The innumerable practices that traditional farmers have developed to attain this goal attest to the importance attributed to microclimate management.

In general, shading causes a reduction of temperature and temperature fluctuations as well as the vapor pressure deficit (VPD) under tropical conditions.

The presence of trees may have both positive and negative overall effects on the water budget of the soil and the crops growing in between or beneath them. The higher soil moisture under trees may have been as a one-way barrier to moisture flow, since it increases the infiltration of rain water while simultaneously reducing evaporation from soil (Wilken, 1972; Muller-Samann, 1986). However, in some situations, especially in semiarid regions, the transpiration of the shade trees may actually increase water stress to the associated crops. Soil temperature i.e., shading tends to exert a buffering effect on temperature fluctuations and extremes.

5.1.1.2. At the tree-animal interface (TAI)

In the above discussion we have seen the positive interaction at the tree-crop interface. Now we will discuss the positive interaction at tree – animal interface. Here you are highly advised to compare and contrast the positive interaction at Tree-crop and Tree animal interfaces.

The positive interactions at the TAI can affect overall system productivity in the following ways:

- First, and most obviously, some part of the autotrophic production that is of no direct use to the farmer (such as weeds or tree fodder) can be transformed into animal biomass with high nutritional and monetary value.
- Secondly, the productivity of the individual system components can be increased, e.g., through the transfer of the individual system components can be increased, e.g., through the transfer of manure as a fertilizer source.
- As with some herbaceous crop plants, animals in the tropics generally benefit from the shade provided by trees. To reduce heat stress, which is one of the main constraints to animal production in the tropics, animals (particularly high-grade, no indigenous animals) tend to seek shade.

Generally, it is evident that shade and high quality fodder are important requirements for better productivity and higher reproduction of animals in the tropics; both can be provided through the inclusion of trees in agricultural systems.

There are only few studies regarding the positive effects of animals on associated trees. Two processes, however, appear to be significant.

- First, animals gathered under shade trees may naturally fertilize the trees through their manure.
- Second, animals may alleviate the competition, e.g., of grass, to which the tree is exposed.

5.2.2. Negative (production-decreasing) interactions

All members of a plant community utilize the same reserves of growth resources such as light, nutrients, water, and CO₂, negative interactions, often through competition, are likely to occur in every plant association (Etherington, 1975; Grime, 1979; Newman, 1983). This competition can

be separated in to that caused by direct interference (real competition), and that caused by exploitation of shared resources which is mediated by other plants or shared predators (apparent competition) (Cannell, 1990). Let us examine the nature of such negative competitions occurring in agroforestry systems.

5.2.2.1. . At the tree-crop interface

The major yield decreasing effects at the TCI arise from competition for light, water, and nutrients, as well as from interactions via allelopathy.

A. Competition for light

Investigations on light interception and competition in agroforestry systems are generally scarce. An additional problem is the difficulty to compare the available results because of the differences in methodologies used in the investigations. However, some insights originate from the few available studies, including some on intercropping of herbaceous species. Shading was found to be more important than below-ground competition in some intercropping studies.

While the availability of light may be the most limiting factor in many situations, particularly those with relatively fertile soils and adequate water availability, the relative importance of light will decrease in semiarid conditions as well as on sites with low fertility soils. Since crops differ in their responses to poor nutrition, competition for light or water may either be reduced or amplified by a shortage of nutrients (Cannell, 1983). A good example of such an interaction between light and nutrients is in the case of cacao (Alvim, 1977). While pod production of cacao is maximum under conditions of high soil fertility and low shade, plants under nutrient stress yield more under shade than in the open; hence the importance of shade trees under low soil-fertility conditions. Generally, the shade tolerance of crop plants depends on the photosynthetic pathway and the product to be harvested. In comparison to leaf yielding plants, fruit and seed yielding crops tend to be relatively shade-intolerant and should therefore be grown in open space where possible (Cannell, 1983). Thus, crops such as coffee, cacao, vanilla, and black pepper, which are traditionally grown under partial tree shade, can be expected to exhibit depressed yields as the intensity of shade increases unless they are subjected to nutrient or water stress.

B. Competition for nutrients

There are innumerable studies indicating how competition for nutrients can reduce crop yields. In most cases, the yield of the agricultural crop is the criterion by which the merit of an agroforestry system is assessed; yield depressions of this component therefore receive more attention than those of the associated tree species. Furthermore, since the crop is usually the smaller component (when compared individually), its root system will usually be confined to soil horizons that are also available to the roots of the trees; but the trees can exploit soil volume beyond reach of the crop. Therefore, the effects of nutrient competition will probably be more severe for the crop components.

C. Competition for water

With the exception of areas with well-distributed rainfall, or azonal sites with a continuous supply of below-ground water, water competition is likely to occur in most agroforestry systems at some period of time; this period may be as short as a dry spell of one or two weeks. The effect of these events depends on the severity of the drought and the drought tolerance of the plants. It also depends on the degree of competition for other resources, especially nutrients. In alley-cropping trials of *Leucaena* with cowpea, castor, and sorghum under semiarid conditions in India, Competition for water appeared more important than shading effects (Singh *et al.*, 1989). Thus, despite the use of drought-adapted plants, water competitions are likely to play a major role in the productivity of agroforestry systems, especially in dry areas.

D. Allelopathy

Allelopathy refers to the inhibition of growth of one plant by chemical compounds that are released into the soil from neighboring plants. A large number of studies have been undertaken in recent years on such allelopathic interactions between plants. Allelopathic properties have been reported for many species, especially trees. Although allelochemicals are reported to be present in practically all plant tissues, including leaves, flowers, fruits, stems, roots, rhizomes, and seeds, information on the nature of active chemicals and their mode of action is lacking. The effects of these chemicals on other plants are known to be dependent principally upon the concentration as well as the combination in which one or more of these substances is released into the environment (Putnam and Tang, 1986).

D. Microclimatic modification for pests/diseases

The effect of plant associations on pest and disease incidence is a potentially important but rather unexplored area. Bacterial and fungal diseases may increase in shaded, more humid environments (Husley and Greenland, 1989). The main reasons for this are probably greater relative humidity and decreased wind, both of which tend to favor fungal growth. However, reduced temperature and humidity fluctuations under shade can also have a suppressing effect on pests and diseases. It seems, then, that the balance between positive and negative effects will have to be assessed for each particular situation.

5.1.2.2. At the tree-animals interface

The most important negative interactions between animals and plants can be classified as:

- Direct effects: Low quality of, or toxic components within, tree fodder can adversely affect livestock production.
- Conversely, mechanical damage of trees or deterioration of soil properties, e.g., through compaction, can have a negative impact on the woody perennial component.

While tree fodder holds great promise, particularly as a dry-season supplement in semiarid areas, its value should not be overestimated. Many species contain secondary compounds that reduce the feed value. The presence of high levels of phenolic compounds (tannins) or strong odors found in the leaves of some plant species may reduce palatability or acceptability of the fodder. In addition, digestibility can be low and the leaves may contain toxins or toxic concentrations of certain micronutrients (Ivory, 1990). The most widely discussed problem is probably that of the toxic compound mimosine found in *Leucaena* fodder. Other particularly harmful compounds include cyanogenic glucosides in *Acacia* species, or robitin in *Robinia* (Ivory, 1990). In certain cases, a toxic or deterrent compound can be extracted and used for pest control, such as azadirachtin in the neem tree (*Azadirachta indica*). Neem leaves, neem oil, and neem “cake” as well as a water extract of crushed seeds, provide a cheap and effective means of pest control (Ahmed and Grainge, 1986; NAS, 1992).

Table 5.2 Toxic or irritant compounds in selected tree fodder species

Species/ Feed	Compound
Acacia	Cyanoglucosides, Fluoracetate, Tannins

Banana leaves	Tannins
Leucaena spp.	Minosine (esp. young leaves, stems and seeds)
Prosopis spp.	Tannins

Source: Adapted from Devendra (1990) and Lowry (1990)

5.2. Component management

Component interaction is an important aspect of agroforestry system. The interaction could be ecological or economical (positive or negative, inter-specific or intraspecific, above ground or belowground). In agroforestry we are not only dealing with the beneficial effects of one component on another, we also deal with the manipulation of negative effects to minimize their influence on the productivity of the overall system. Therefore, component management is an important part of agroforestry sciences.

There are different factors that are governing interaction between components in agroforestry. The magnitude of interactive effects between trees and other components of agroforestry systems depends on the characteristics of the species, their planting density, and spatial arrangement and management of the trees.

Tree management:

There are certain management techniques which are applied to trees and shrubs in agroforestry systems. Some of these techniques are similar to those used in the management of trees in forestry plantations, but others are different. The most important management techniques with regard to the part of the tree which is above ground are: Pruning, Lopping, Pollarding, Coppicing, Thinning.

Pruning

Removal of branches from the lower part of the tree crown is known as pruning or side pruning. While pruning a tree, branches are always cut near the stem.

The objectives of pruning in agroforestry are threefold:

- Reduction of shade for crops near the tree
- Improving the quality of the trunk, mainly for timber and poles
- Early harvest of branch wood for fuel or other use.

- Removal of live or dead branches or multiple leaders from standing trees for the improvement of the tree or its timber.
- It allows the grower to manipulate the growth and development of the trees left after thinning to improve the quality of the tree and to increase agriculture returns
- Natural; natural death and fall of branches of standing trees grown closely due to deficiency of light or decay etc.
- Artificial: Removal of branches with sharp tools in a dense crop.
- Pruning lower branches close to the trunk of tree makes small knotty core which gives clear straight grain timber.
- Removal of too many branches will retard the growth
- If pruning is done too late, the central core of knotty wood become large thus reducing value of tree

Too much pruning may reduce the growth of certain species. For young trees, at least four or five layers of the green branches should remain uncut, while older trees of certain species can tolerate more severe pruning. Pruning should be done at least up to the height the adjacent crops if trees are growing in fields. Such pruning facilitates farming operations and reduces competition. The best time for pruning is towards the end of the dry season when the work will not interfere with growing crops and when the workload in other agriculture tasks is not so heavy.

Lopping

Lopping is distinguished from pruning in that **branches are not cut from the base**. Also lopping is not always done starting from the lower part of the tree but can be more haphazard. If any selection of branches is made, the main criterion is often a good green leafy biomass since the lopping is usually done to obtain branches for fodder. One of the main advantages with this technique is that it allows harvest without killing the tree. All tree species can be lopped, but the growth rate of certain species can be retarded if they are heavily lopped.

Pollarding

If **all the branches and the top part of a tree are cut off** this is known as pollarding. There can be several objectives with pollarding:

- Early harvest of wood, fodder or other biomass
- Production of wood or fodder that is out of the reach of livestock, hence there is no need for protection from browsing
- Reduction of shade for crops near by
- Regeneration of the tree crown to promote growth of the trunk for timber or poles.

Some commonly pollarded species are:

- *Balanites* spp., *Bridelia micrantha*, *Casuarina* spp., *Cordia abyssinica*, *Croton* spp., *Erythrina abyssinica*, *Faidherbia (Acacia) albida*, *Ficus sycomorus*, *Grevillea robusta*, *Jacaranda mimosifolia*, *Manihot glaziovii*, *Markhamia lutea*, *Morus alba*.

Coppicing

Many species of trees and shrubs have the ability to resprout after the whole tree has been cut. If this ability is utilized for regeneration of the tree the practice is known as coppicing. Coppicing can almost be regarded as a method of tree propagation since it can substitute for the task of planting a new tree after a mature one is felled. Systematic coppicing is applied as the management technique in alley cropping, and it may be an option for trees on soil-conservation structures. In such a situation coppicing may be done annually, but in other situations, e.g. regeneration of *Eucalyptus* for pole production, it may be much less frequent. In that case, an interval of 6-8 years may be more suitable. Not all tree species will coppice after being cut. Some commonly coppiced species are:

- *Calliandra calothyrsus*, *Cassia siamea*, *Cassia spectabilis*, *Eucalyptus* spp., *Leucaena leucocephala*, *Markhamia lutea*. Certain species coppice well when young but may not do so if cut at maturity. *Casuarina* spp., *Crevillea robusta*, *Sesbania sesban* and some *Albizia* spp.

Thinning

Trees established by direct seeding or that have been planted with little space between them will soon start to compete with each other. A dense stand initially promotes straight growth and small branches, but later the trees must be thinned otherwise they will grow too slender and eventually not reach the desired size. Thinning is particularly important for trees grown in woodlots, but applies also to other situations where trees are growing close to each other. Thinning can, for example, be done by removing every second tree or two out of every three trees. Thinning is also a way of obtaining some early harvest. Thinning is mainly done:

- ❖ To improve the hygiene of the crop by removing dead, dying and diseased trees
- ❖ To ensure best physical conditions of growth
- ❖ To obtain a desired type of crop
- ❖ To afford protection from the spread of insects and diseases
- ❖ To improve the quality of wood

- ❖ Increase the net yield and financial return from the crop

CHAPTER 6. AGROFORESTRY AND SOILS

6.1. Beneficial effects of trees on soils

It is generally believed that trees and other vegetation improve the soil beneath them. There are tremendous evidences that demonstrate the role of trees in conserving and improving the soil fertility. There are different mechanisms by which trees are improving soil fertility. For easy understanding, we grouped them under three main categories:

1. Process which **increase addition to the soil**: Trees can add nutrients in various ways in to the soil system. Among these are:

- a) Maintenance of soil organic matter:

“Agroforestry systems can maintain soil organic matter and biological activity at levels satisfactory for soil fertility”. This is one of the hypothesis in agroforestry. The effect of trees in maintaining soil organic matter (OM) levels through the supply of litter and root residues are a major cause of soil fertility improvement. It is the prime mover, from which stem many other soil improving processes. Conversely, decline in soil OM invariably accompanies land degradation. Soil OM under natural ecosystems is set primarily & by the clay content and types of clay minerals; it is lower for sandy soils, highest for 2:1 lattice clays. In agricultural and agroforestry ecosystems, soil management becomes a major variable, normally reducing OM in rain-fed agriculture by up to 50% and in degrading systems by more. Dense, mixed spatial agroforestry systems maintain soil organic matter, in some cases as equal as natural vegetation. But, organic matter production reduced in case of reclamation agroforestry and in open spatial systems. In hedgerow intercropping, a clear differential develops after some years between organic matter under the hedgerows and in the alleys, visible in the top soil colour if the hedgerows are removed.

- b) Nitrogen fixation

Nitrogen fixation is a process by which atmospheric nitrogen is converted in to nitrate by the action of microorganisms. The biological nitrogen fixation takes place by non-symbiotic and symbiotic menses.

Non symbiotic nitrogen fixation:

None symbiotic nitrogen fixation is carried out by soil microorganism not dependent on plants. In this type of nitrogen fixation the bacteria's are living freely in the soil without establishing any symbiotic association with plants. Non symbiotic fixation is important in natural ecosystem but the amount of nitrogen fixed are small in relation to the greater requirements to agro- ecosystems. Examples of trees without symbiotic association are Casuarinas and Other species belonging to the casuarinaceae.

Symbiotic nitrogen fixation

Unlike non symbiotic nitrogen fixation, symbiotic nitrogen fixation occurs through associations of plant roots with N- fixing bacteria. The symbioses could take place in two ways:

1. Between leguminous species and Rhizobium or Bradyrhizobium forming nodules on roots
2. Between a small number of non-leguminous genera and Frankia bacteria. Examples of trees with symbiotic N-fixation. *Leucaena leucocephala*, *Sesbania sesban*, *Albizia* spp, *Prosopis* spp and Numerous *Acacia* species eg. *Faidherbia albida*.

Biological nitrogen fixation added nitrogen into the soil plant system at no cost. The biological nitrogen fixation has the following advantages:

- a. There is no fertilizer cost or minimum fertilizer is used
- b. Loss from the atmosphere is negligible. e.g., *Leucaena leucocephala*, if well established as pure stand can fix 100-500kg N/ha/year. *Sesbania rostrata* in swamp rice system it can fix 500kg N/ha/year. If current price of UREA is 600birr/quintal, it will be possible to save 600 to 3000 birr by application of *Leucaena leucocephala* in agroforestry system. Nitrogen is transferred from the fixing to non fixing species in three forms
 - a. Below ground, directly plant to plant
 - b. Below ground, via the soil
 - c. Above ground, via litter and pruning

Direct plant to plant transfer is referring to exudation of growth promoting substances by the rhizosphere of trees. In this type of transfer the root system of the two plants have direct physical contact. Transfer via the soil takes place through dropping of nodules (small swellings in the root system of plants where the nitrogen fixing bacteria's are functioning) and roots. But, generally, the main transfer is via above ground residues. The quantity of nitrogen fixed do not differ as

between good herbaceous and woody nitrogen fixing species. However, the difference may arise based on:

- The time of application
- The manner of application to the soil

The choice either to use woody plants or herbaceous plants is made on the ground (based on the management preference of the user), not intrinsic

c) Nutrient uptake

In addition to nitrogen fixation, two processes have the potential to increase nutrient inputs to the plant soil system.

A. Nutrient retrieval from weathering rock-Part material

B. Increased inputs from rain and dust- Atmospheric input

Plants (trees) take nutrients from deep soil horizons and weathering rock.

The recycling hypothesis states that, by the inclusion of trees, agroforestry system can achieve a condition intermediate between these extremes, by

- ⇒ Reducing losses
- ⇒ Increasing plant uptake

In agroforestry, generally there are:

1. Gains and loss
2. Uptake and return

Gain and losses

Gain and losses represents nutrient addition to the system or nutrient lost from the system. Gain and loss are processes external to the system. The followings are nutrient gains to the system from external sources.

- ❖ Atmospheric input (rain and dust)
- ❖ Fertilizer
- ❖ Organic addition (compost, manure etc)
- ❖ Symbiotic fixation (nitrogen)
- ❖ Non symbiotic fixation (nitrogen)
- ❖ Rock weathering (excluding N)

Nutrients are lost from the system to the external system through the following mechanisms:

- Erosion, leaching, harvest, gaseous losses eg., volatilization and burning (nitrogen, sulphur).

Uptake and return

Unlike gain and loss, uptake and return are internal transfers, respectively soil to plant and plant to soil.

Uptake: uptake of nutrients by plants from the soil

Return : includes the following

- Mineralization of litter
- Mummification of litter
- Ash from burning

Now onwards we will discuss how to calculate nutrient balance and nutrient cycling ratio in the plant soil system. For any specific nutrient:

Nutrient balance (kg/ha/yr) = Gains – losses

The results of nutrient balance could be negative or positive. The negative result indicate the deficiency of a particular nutrient in the soil and the positive result indicate the availability of a particular nutrient in a given soil.

Nutrient cycling ratio (%) = $\frac{\text{uptake} + \text{Return}}{\text{Gain} + \text{losses} + \text{uptake} + \text{Return}} \times 100$

Generally, under closed natural system the nutrient cycling ratio is approaching 100%. This is quite common to old natural forests and woodlands. Where there is no internal recycling (the net result of uptake and return is zero) the nutrient cycling ration will become 0 (zero). In agroforestry system (rotational system) there will be an average nutrient cycling ratio. In a well managed agroforestry system, especially in a system involving different plant species and arrangement, it is possible to have a nutrient cycling ratio approaching to the natural ecosystem.

Exercise. 1

- 1- What will happen if the nutrient balance is zero?
- 2- What measures are needed if the nutrient balance is negative?

Example 1- phosphorus cycle in a rain forest in Latin America showed that (kg/ha/yr)

Gain = 1.0 kg

Losses = 0.9 kg

Uptake = 11.0 kg and,

Return = 11.9 kg

a) Calculate the nutrient balance?

b) Calculate the phosphorous cycling ratio?

Nutrient balance = Gain – Loss

$$= 1.0 \text{ kg} - 0.9 \text{ kg}$$

$$= \mathbf{0.1 \text{ kg}}$$

Nutrient cycling ratio = $\frac{\text{uptake} + \text{Return}}{\text{Gains} + \text{losses} + \text{uptake} + \text{Return}} \times 100$

$$= \frac{11.0 + 11.9}{1.0 + 0.9 + 11.0 + 11.9} \times 100$$

$$= \frac{22.9}{24.8} \times 100$$

$$= \mathbf{92 \%}$$

$$24.8$$

$$= \mathbf{92 \%}$$

Example-2. Nitrogen flow in your working area showed that 100kg fertilizer added to the system and 50kg nitrogen returned into the system after the decomposition of plant roots. However, 50 kg nitrogen leached out and 100kg lost through harvest. The study further showed that 150kg nitrogen was taken by plants for their metabolic processes. Calculate the nitrogen balance and nitrogen recycling ratio? Given:

a- gain= 100kg – as fertilizer

b- losses= 100kg (harvest) and 50kg (leaching)

c- uptake = 150 kg (uptake by plants)

d- return = 50kg (root residue)

Nutrient balance = Gain – Loss

$$= 100 - 150$$

$$= \mathbf{- 50 \text{ (negative)}}$$

Nutrient cycling = $\frac{\text{uptake} + \text{Return}}{\text{Gain} + \text{loss} + \text{uptake} + \text{Return}} \times 100$

$$= \frac{50 + 150}{100 + 150 + 150 + 50} \times 100$$

$$100 + 150 + 150 + 50$$

$$= \mathbf{44 \%}$$

Exercise.2

1- A group of soil scientists conducted a detail study on nitrogen balance and nitrogen cycling ratio. In their study, scientists found that 60kg of nitrogen is leached down and becoming inaccessible by the root system of plants; and additional 50kg eroded from the site. However, 100kg atmospheric nitrogen is fixed by leguminous plants and 150kg nitrogen added in the form of Urea (artificial fertilizer). Scientists further identified that 80kg of nitrogen is used by plants for their metabolic processes and 40kg nitrogen released to the system after decomposition of the litter.

A- Identify the givens

B- Calculate the Nitrogen balance

C- Calculate the Nitrogen recycling ratio

CHAPTER 7. AGROFORESTRY POLICES FOR SUSTAINABLE LAND USE

Agroforestry has long been recognized for its potential as a stable and sustainable production system and for its potential contribution to broader agricultural systems. Agroforestry can extend the amount of time that a given area can be productive, improve livelihoods, and contribute to forest and biodiversity conservation. Constraints to agroforestry systems include high interest rates, unclear institutional responsibilities, limiting policy frameworks, poorly developed markets, and inadequate research and extension. Support to agroforestry needs to address this broad range of issues, within flexible program designs. In this chapter we will discuss the policy issues.

7.1. Benefits of Agroforestry that Attracts the Attention of Policy Makers

7.1.1. Benefits of agroforestry

Poverty reduction

Agroforestry can reduce poverty directly by providing lumber, fuelwood, fruit and nuts, and livestock fodder, all of which can be sold to generate income or fulfill basic family needs. The sale of timber is particularly important to poverty reduction, as returns are long term. Indirectly, agroforestry can increase crop production and incomes through conservation of soil and soil moisture.

Reducing vulnerability

It is well understood that agroforestry has long been a traditional coping mechanism to reduce production-related risks, particularly during times of drought or crop failure. Trees store biomass during good production seasons and, when annual crops fail, can be harvested to provide income to purchase food and other needs.

Environmental sustainability

Agroforestry is associated with positive environmental outcomes because of the role trees play in larger ecosystem functions. Trees can improve soil quality in various ways: root systems prevent soil erosion, leguminous species fix nitrogen and improve nutrient recycling, and detritus (debris) from trees increases the organic content of soil. Well-developed agroforestry systems provide habitat for wild animals and contribute to biodiversity. Agroforestry affects climate change by storing carbon and offsets deforestation by providing an alternative source of wood products. Finally, agroforestry can act as a buffer between protected forests and surrounding agricultural land and minimize edge effects in natural forests.

Contributions of Agroforestry to Land Use Sustainability

Despite advanced technical knowledge, many land users today continue to focus on current production and not on protecting the natural resources base on which they and future generations will depend. Often this is because commonly-used technologies and production systems focus on quick production responses rather than long-term effects. However, agroforestry largely evolved with sustainability concerns (resiliency, diversity, and avoiding negative side effects) in mind. It blends production with protection, and focuses on a holistic approach to land management. Fuelwood, poles, lumber, and fodder are agroforestry tree products often sorely needed by rural people. Trees also moderate the microclimate, protect the soil, provide shade to livestock, and enhance crop production by reducing wind effects on crops. At the same time, the role of trees in agricultural systems is not always positive. For example, trees can compete with crops for space, light, nutrients, and water, reducing the overall value of both trees and crops. These problems require experience and careful study to find combinations of trees, crops, and livestock that maximize overall production. In some cases, this can mean no trees; in other cases, it can mean total forest cover. Agroforestry is in between. Policies should encourage the most productive uses of land and find positive ways to introduce trees.

7.2. Policy Measures to Encourage Appropriate Agroforestry

It is generally believed that policy can encourage agroforestry to improve existing unsustainable land use practices. Because agroforestry represents an array of possible land uses, policies in many sectors can affect it. For example, price supports for agricultural crops, subsidies for fertilizer and other agricultural inputs, low interest credit to farmers, and investment in agricultural education and research can also affect agroforestry. Similarly, policies that favor certain types of land use, can affect agroforestry. The differences in these types of policies are mainly a matter of degree. However, clearly, some policy measures particularly affect agroforestry and the use of trees in land-use systems. Below are examples of such measures that policymakers need to review.

7.2.1. Policy and Implementation Issues

7.2.1.1. Land tenure.

Agroforestry requires a long-term investment, with benefits delayed by three years or more, farmers must be willing and able to invest in medium-to long-term land improvements. Land tenure systems that do not guarantee continued ownership and control of land are not likely to be conducive to the adoption of long term strategies (and relatively short term practices that include benefits which will only be realized in the long run) such as agroforestry. Secure land rights, in particular, have proven pivotal in determining whether the benefits of agroforestry reach the intended beneficiaries. Where land tenure is not legally established, landholders will seldom commit to long-term land improvements. Land tenure issues are complex, but they must be addressed if agroforestry is to be promoted as an investment.

7.2.1.2. Market access

Direct and immediate income that can be derived from a land use system will be an important criterion in the appraisal of the social acceptability of an agroforestry. Agroforestry's contribution to poverty reduction is dependent on people's access to forest product markets. Market access can be improved through construction of roads, development of farmer organizations to increase the bargaining power of producers, or negotiation of contracts between farmers and larger forestry companies. Market analysis is essential to assess suitability of tree species as cash crops. Products from trees on farmers are considered as free goods in many farming societies. Creating marketing opportunities for these free goods, and thus increasing the demand for these products, will also

require making appropriate provisions for meeting the local need of such locally produced and freely available items

7.2.1.3. Gender

In the many situations where it is women's responsibility to gather fuelwood and tend household gardens, agroforestry can give women more control over a productive resource. In other cases, where men exert traditional ownership rights over trees, they may not have control over the use of agroforestry products or proceeds from their sale. Gender issues, and the site-specific factors that affect them, should be addressed by gender assessments to ensure equitable access to agroforestry benefits.

7.2.1.5. Participation of poor people

Care needs to be taken when designing an agroforestry activity to ensure that vulnerable groups are not excluded, especially the landless poor and people with poorly defined land rights. Inclusive implementation mechanisms need to be incorporated into the activity to enable the sharing of benefits and the encouragement of democratic approaches to decision making in the community.

7.2.1.6. Regulatory and Legal Policies

There are a variety of these types of policies that decision makers can use:

- Tenure laws sometimes make trees the property of the state. In such cases, farmers have "no" incentive to plant trees, no matter how useful they would be. Such laws are often a carry-over from government ownership of forest lands and attempts to control forest clearing. Policymakers should review these laws carefully and revise them where appropriate without jeopardizing remaining forests.
- Laws and regulations that prevent farmers from harvesting farm trees should be reviewed to increase incentives to plant and manage trees as part of the total farm enterprise.
- Laws or regulations that control public forest management and use can affect the incentive for farmers to grow trees on their own land as these laws affect markets or the availability of free or subsidized wood. While policymakers have to consider such laws and regulations in a much broader context than agroforestry, they need to remember the impacts on farmer tree-growing activities when planning public forest regulations.
- Some countries, such as Japan, for many years have had social systems that regulate water use. Downstream land and water users pay upstream land users for soil and water

conservation practices that affect the downstream land uses. Legal mechanisms can ease these negotiations between upstream and downstream land users and encourage agroforestry practices.

7.2.2. Fiscal (Tax and Subsidy) Policy Measures

Policymakers can use several types of fiscal policies. These fiscal measures are presented as follows:

- ✚ Sometimes, governments apply (formally and informally) special taxes to tree harvesting on private land. This discourages tree planting. Policymakers must consider taxes in the context of broader objectives of promoting sustainable land use.
- ✚ Governments often subsidize farmers who produce tree seedlings. These subsidies can help get local communities into small-scale nursery production and give farmers a readily accessible, cheap source of planting stock.
- ✚ Subsidies that affect the use of fuels, such as kerosene, electricity, and fuelwood, can also encourage or discourage farm planting and managing trees for fuelwood.
- ✚ Governments can levy taxes on downstream land and water users to generate revenues for upstream soil and water conservation projects.

CHAPTER 8. DIAGNOSIS AND DESIGN METHOD

8.1. Definition and Concepts

Diagnosis and design is a methodology for diagnosis of land management problem and design of agroforestry solution. Diagnosis and Design (D&D) is a systematic and objective methodology developed by International Council for Research in Agroforestry (ICRAF) to initiate, monitor, and evaluate agroforestry programs. D&D is based on the philosophy that knowledge of the existing situation (diagnosis) is essential to plan and evaluate (design) meaningful and effective programs in agroforestry research for development. The methodology plays a strategic role in all the phases of the agroforestry research process. It borrows from other methodologies used by research or development agencies, such as baseline surveys and feasibility studies. However, D&D is unique in that it has been specially developed for the following purposes (Raintree, 1987):

- to describe and analyze existing land use systems
- to design appropriate agroforestry technologies to alleviate those constraints;
- to design appropriate research work, such as trials and funkier surveying.

The basic unit of D&D analysis is the land use system (LUS). The LUS can be defined and analyzed at the level of a country, ecozone, farming system, crop system, or any other unit. The structure and function of any LUS are determined by climatic, physical, biological, technological, economic, social, and political factors. D&D focuses on the interactive effects these factors have on the LUS, and searches for opportunities for improved system development in the LUS. For the purposes of a D & D exercise a **land use system** is defined as a **distinctive combination** of three inter related factors: the **land resources** exploited by a particular **technology** to satisfy the production objectives of a particular type of **land users**. This definition contrasts with other common by used definitions, which do not formally recognize the **land user** (i.e. the human element) as part of the system. The key features of D & D approach are:

Flexibility: D & D is a flexible discovery procedure which can be adapted to fit the needs and resources of different users.

Speed: D & D has been designed with the option of a “rapid appraisal “application at the planning stage of a project with in depth following during project implementation.

Repetition: D & D is an open-ended learning process. Since initial designs can almost. Always be improved the D & D process needn’t end until further improvement are no longer necessary.

The systematic elaboration and adaptation of this basic logic (i.e. D & D) to agroforestry has been developed with three main users in mind;

- ⇒ Researchers
- ⇒ Extension agents
- ⇒ Community development fieldworkers

8.2. Diagnosis and Design implementation

D&D can be done at two levels:

1. Macro D&D
2. Micro level

Macro D&D is a large-scale analysis of an ecozone within a country or a group of countries. For example, ICRAF has conducted collaborative macro D&D in Uganda, Rwanda, and Burundi for the bimodal highland ecozone. Macro D&D is important for deciding on national agroforestry research-and extension agenda at-the national level.

Micro D&D, in contrast, focuses on one land use system (LUS) within the larger ecozone that has special priority for agroforestry intervention. Micro D&D involves a detailed analysis of

households and production systems in the LUS. It leads to guidelines for research that will address the constraints of the prioritized LUS. The objectives of micro D&D are similar to those of macro D&D. The major difference is that whereas macro D&D has a broad scope (i.e., an ecozone), micro D&D focuses on detailed analysis of one prioritized LUS. The three main objectives of micro D&D are:

- ✓ to describe and analyze an LUS in order to identify its constraints; and
- ✓ to design and evaluate agroforestry technologies to address the constraints
- ✓ to design and evaluate appropriate research programs aiming to develop these technologies.

Since resources are inevitably limited, a country, institution or project will have to be selective in deciding which LUSs should be subjected to micro D&D. The choice of the LUS for micro D&D depends on criteria such as:

- ⇒ political and economic importance of the system,
- ⇒ technical potentials for improvement of the LUS, and
- ⇒ scientific expertise and other resources in the national collaborating institutions for carrying out research in the LUS.

Although it is not essential for a macro D&D exercise to precede a micro D&D exercise, the task of prioritizing LUSs, defining the research focus, and defining areas for institutional collaboration will be much easier after macro D&D has been completed.

8.2.1. Analysis of Land Use System and Constraints

This phase of micro D&D aims at:

- prioritizing the needs of the household;
- identifying production constraints (both those that can and cannot be manipulated); and
- assessing potentials for system development.

The basic framework used for this analysis is a farming system, where the decision-making unit is the household. The household usually manages a combination of crop, livestock, and tree production systems, along with other non-agricultural and off-farm activities, to satisfy its basic felt needs of food, cash, fuelwood, building materials, and security. Besides endogenous factors, the farming system is influenced by exogenous factors of a political, social, economic, or technological nature. Understanding the interactions within the farming system and the effects of environment is essential for prioritizing the needs of the household, identifying production constraints, and assessing potentials for system development. The micro D&D research team,

therefore, will want to quantify resources, management, and yield of each component of the farming systems, including characteristics and priorities of the household.

8.2.2. Description of Land Use Systems (LUS)

All delineated LUSs are described by specifying the characteristics that are known to affect their current management and performance, and would be expected to affect the introduction of potential agroforestry technologies. These characteristics are outlined below:

- I. Title of system
- II. Location: administrative and political divisions, with map if available.
- III. Ecological characteristics: This includes
 - Agroecological zone,
 - Altitude range (m),
 - Topography: slope (gradient),
 - Rainfall: total annual, monthly levels, range,
 - Number of growing seasons: with months and growing days,
 - Soil: type, texture, pH, fertility, etc,
 - Hydrology: river network, water table level, etc., and
 - Vegetation: natural and secondary
- IV. Socioeconomic characteristics
 - Total land area in the LUS (km² or ha
 - Total population in the LUS,
 - Population density: persons/km²,
 - Ethnic groups: religion, culture, etc.,
 - Tenure system: ownership or user rights for crops pastures, land, trees,
 - Farm income: levels and sources, and
 - Infrastructure: roads, electricity, commercial centers, etc.
- V. Land use characteristics
 - ✓ Farm size: average, range, distribution,
 - ✓ Spatial arrangement: location of homestead, crop, livestock, trees,
 - ✓ Major and minor agricultural production activities: food and cash crops, large and small livestock, trees, etc.,
 - ✓ Area covered by various components: hectare
 - ✓ Crop production: main crops - land preparation, planting methods/timing, use of manure/fertilizer/pesticides, weeding, soil conservation, harvesting, level of production, storage, etc.,
 - ✓ Livestock production: type, herd size, breeds, feed sources and management (tree/zero grazing, etc), type/use of output.
 - ✓ Tree production: species, main uses, land use niche, management, and arrangement, etc; traditional and new agroforestry systems (Note arrangement, management, or any data on performance,
 - ✓ Production systems interactions: relationship between crops, livestock, trees.
- V. Resources/supporting services
 - Labor availability and utilization: family owned, hired, communal, etc.,
 - Farm power and equipment used,
 - Marketing: markets, marketing channels, prices, etc.,
 - Credit facilities: type available and for what farm activities,

- Extension services: nature and organization of extension services, and
 - Local organization: cooperatives, farm organizations, churches.
- VI. Development activities and policies
- ❖ Review of relevant government policies and strategies, and
 - ❖ Review of research and development projects, e.g., in agroforestry.

8.3. Basic procedures in conducting D and D

The basic procedures of D & D consist of five stages as described in the table below

Table 8.1. Basic procedure of the D and D Methodology (Raintree 1987)

D & D stages	Basic questions to answer	Key factors to consider	Mode of inquiry
Pre-diagnostic	Definition of the land uses system and site selection (which system to focus on?).	Distinctive combinations of resources, technology and land user objectives	Seeing and comparing the different land use systems
Diagnostic	How does the system work? (How is it organized how does it function to achieve its objectives)	Productive objectives and strategies, arrangement of components.	Analyzing and describing the system.
	How well does the system work (what are its problems, limiting constraints, problem generating syndromes & intervention points?)	Problems in meeting system objectives production short falls, sustainability problems	Diagnostic interviewer and direct field observations.
Design and evaluation	How to improve the system? (What is needed to improve system performance?)	Causal factors constraints and intervention points.	Troubleshooting the problem subsystem.
Planning	What to do to develop and disseminate the improved system	Specification for problem solving or performance enhancing interventions	Iterative design and evaluation of alternative
Implementation	How to adjust to new information.	Research and development needs, extension needs	Research design project planning
		Feedback from on-station research on –farm trials and special studies	Rediagnosis and redesign in the light of new information